III B.Tech II Semester

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23A31603a	**GRAPH NEURAL NETWORKS (Professional Elective-II)**	**3**	**0**	**0**	**3**

Course Objectives:

- To introduce the fundamentals of graph theory and graph-structured data.
- To explore the concepts of neural networks extended to non-Euclidean domains.
- To understand architectures and algorithms behind various types of GNNs.
- To apply GNN models in real-world applications such as recommendation, social networks, and bioinformatics.
- To enable students to build and evaluate GNN models using frameworks like PyTorch Geometric and DGL.

Course Outcomes:

Upon completion of the course, students will be able to:

- Understand the basics of graph structures and their significance in machine learning.
- Learn and implement different types of GNN architectures.
- Apply GNNs to real-world structured data problems.
- Use modern libraries and tools to train and evaluate GNNs.
- Analyze the effectiveness and limitations of GNNs in different domains.

UNIT I: Fundamentals of Graph Theory and Machine Learning on Graphs
Introduction to Graphs: Nodes, Edges, Adjacency Matrix, Types of Graphs: Directed, Undirected, Weighted, Bipartite, Graph Traversal Algorithms (BFS, DFS), Graph Representations for ML (Adjacency List, Matrix, Laplacian), Node, Edge, and Graph-level Prediction Problems, Motivation and Challenges for Learning on Graphs.

UNIT II: Spectral and Spatial Methods for Graph Learning
Spectral Graph Theory Basics, Graph Convolution via Spectral Methods, Chebyshev and First-order Approximations, Spatial Graph Convolutions, Comparison of Spectral vs Spatial GNNs, Graph Laplacian and Eigenvalue Properties.

UNIT III: Graph Neural Network Architectures
Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), GraphSAGE: Sampling and Aggregation, Graph Isomorphism Networks (GIN), Message Passing Neural Networks (MPNNs), Inductive vs Transductive GNN Learning.

UNIT IV: Applications of GNNs
Node Classification (e.g., Cora, Citeseer), Link Prediction (e.g., Recommender Systems), Graph Classification (e.g., Molecule Property Prediction), Traffic Forecasting and Social Network Modeling, GNNs in Healthcare and Bioinformatics, Explainability and Interpretability in GNNs.

UNIT V: Implementation, Optimization, and Recent Advances
Overview of PyTorch Geometric and DGL, Data Loading and Preprocessing for Graph Datasets, Model Training, Loss Functions, and Evaluation Metrics, Hyperparameter Tuning in GNNs, Recent Research Trends and Architectures (e.g., Heterogeneous GNNs, Graph Transformers), Challenges and Future Directions in GNNs.

Textbooks:

1. Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, Philip S. Yu, *A Comprehensive Survey on Graph Neural Networks*, IEEE Transactions on Neural Networks and Learning Systems, 2021.
2. Yao Ma, Jiliang Tang**,***Deep Learning on Graphs*, Cambridge University Press, 2021.
3. William L. Hamilton, *Graph Representation Learning*, Morgan & Claypool Publishers, 2020.

Reference Books:

1. Barrett, Jure Leskovec, *Mining of Massive Datasets*, Cambridge University Press.
2. Thomas Kipf**,***GCN and related papers and tutorials* (arXiv).
3. Petar Veličković, *Graph Attention Networks* (original paper and slides).
4. Michael Bronstein et al.**,***Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges* (arXiv preprint).

Online Learning Resources:

1. **https://pytorch-geometric.readthedocs.io/**– PyTorch Geometric Docs
2. **https://cs.stanford.edu/people/jure/** – Stanford GNN Projects
3. **https://www.coursera.org/learn/graph-neural-networks** – Coursera GNN Course by Stanford